THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2346	digit sum 15 is a multiple of 3
2	**(a)** 4527	digit sum 18 is a multiple of 9
3	**(b)** 3846	last digit 6 is even
4	**(c)** 21	Add all the digits
5	**(c)** even	Odd + odd = even
6	**(d)** 2743	Write the digits in the opposite order
7	**(b)** 4554	A palindrome reads the same from both ends
8	**(a)** 12	Smallest common multiple
9	**(b)** 8205	digit sum 15 is a multiple of 3
10	**(d)** 6318	digit sum 18 is a multiple of 9

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Yes (1); digit sum 15 is a multiple of 3 (1).	1+1
12	Known digit sum $= 19$ (1); smallest digit $= 8$ (1).	1+1
13	Multiples of 3: 3, 6, 9, 12 (1); LCM $= 12$ (1).	1+1
14	No (1); digit sum 17 is not a multiple of 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 2: Yes, last digit 0 is even (1); by 5: Yes, last digit 0 is 0 or 5 (1); by 10: Yes, last digit 0 is 0 (1).	1+1+1
16	Digit sum of the other digits $= 7$ (1); need a total that is a multiple of 3 (1); digits: 2, 5, 8 (1).	1+1+1
17	12 and 16: common factors give HCF $= 4$ (1.5); LCM $= 48$ (1.5).	1½+1½
	OR: By 2: No, last digit 5 is not even (1); by 5: Yes, last digit 5 is 0 or 5 (1); by 10: No, last digit 5 is not 0 (1).	1+1+1
18	Digit sum of the other digits $= 15$ (1); need a total that is a multiple of 3 (1); digits: 0, 3, 6, 9 (1).	1+1+1
	OR: 18 and 24: common factors give HCF $= 6$ (1.5); LCM $= 72$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	By 2: Yes, last digit 0 is even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: Yes, last digit 0 is 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: Yes, last digit 0 is 0 (1);	1+1+1+1+1
	OR: Remainder when 4537 is divided by 9 is 1 (2); number to add $= 9 - 1 = 8$ (2); new number 4545 (1).	2+2+1
20	By 2: No, last digit 5 is not even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: Yes, last digit 5 is 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: No, last digit 5 is not 0 (1);	1+1+1+1+1
	OR: Remainder when 6182 is divided by 9 is 8 (2); number to add $= 9 - 8 = 1$ (2); new number 6183 (1).	2+2+1

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5712	last two digits 12 are divisible by 4
2	**(b)** 17	Add all the digits
3	**(a)** odd	Even + odd = odd
4	**(c)** 1805	Write the digits in the opposite order
5	**(b)** 7227	A palindrome reads the same from both ends
6	**(a)** 18	Smallest common multiple
7	**(c)** 6132	digit sum 12 is a multiple of 3
8	**(c)** 7254	digit sum 18 is a multiple of 9
9	**(a)** 8365	last digit 5 is 0 or 5
10	**(c)** 19	Add all the digits

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Known digit sum $= 13$ (1); smallest digit $= 5$ (1).	1+1
12	Multiples of 4: 4, 8, 12, 16, 20 (1); LCM $= 20$ (1).	1+1
13	Yes (1); digit sum 18 is a multiple of 3 (1).	1+1
14	Known digit sum $= 8$ (1); smallest digit $= 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 2: Yes, last digit 2 is even (1); by 5: No, last digit 2 is not 0 or 5 (1); by 10: No, last digit 2 is not 0 (1).	1+1+1
16	Digit sum of the other digits $= 15$ (1); need a total that is a multiple of 3 (1); digits: 0, 3, 6, 9 (1).	1+1+1
17	20 and 30: common factors give HCF $= 10$ (1.5); LCM $= 60$ (1.5).	1½+1½
	OR: By 2: Yes, last digit 0 is even (1); by 5: Yes, last digit 0 is 0 or 5 (1); by 10: Yes, last digit 0 is 0 (1).	1+1+1
18	Digit sum of the other digits $= 13$ (1); need a total that is a multiple of 3 (1); digits: 2, 5, 8 (1).	1+1+1
	OR: 14 and 35: common factors give HCF $= 7$ (1.5); LCM $= 70$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	By 2: Yes, last digit 6 is even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: No, last digit 6 is not 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: No, last digit 6 is not 0 (1);	1+1+1+1+1
	OR: Remainder when 7345 is divided by 3 is 1 (2); number to add $= 3 - 1 = 2$ (2); new number 7347 (1).	2+2+1
20	By 2: Yes, last digit 0 is even (1); By 3: No, digit sum 11 is not a multiple of 3 (1); By 5: Yes, last digit 0 is 0 or 5 (1); By 9: No, digit sum 11 is not a multiple of 9 (1); By 10: Yes, last digit 0 is 0 (1);	1+1+1+1+1
	OR: Remainder when 2918 is divided by 5 is 3 (2); number to add $= 5 - 3 = 2$ (2); new number 2920 (1).	2+2+1

Mistakes to avoid:

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- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** odd	Odd x odd = odd
2	**(a)** 4629	Write the digits in the opposite order
3	**(c)** 12921	A palindrome reads the same from both ends
4	**(d)** 30	Smallest common multiple
5	**(a)** 5472	digit sum 18 is a multiple of 3
6	**(d)** 5238	digit sum 18 is a multiple of 9
7	**(a)** 7240	last digit 0 is 0
8	**(d)** 21	Add all the digits
9	**(a)** even	One even factor makes the product even
10	**(b)** 5137	Write the digits in the opposite order

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Multiples of 6: 6, 12, 18, 24 (1); LCM $= 24$ (1).	1+1
12	No (1); digit sum 19 is not a multiple of 3 (1).	1+1
13	Known digit sum $= 20$ (1); smallest digit $= 7$ (1).	1+1
14	Multiples of 9: 9, 18, 27, 36 (1); LCM $= 36$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By 2: No, last digit 5 is not even (1); by 5: Yes, last digit 5 is 0 or 5 (1); by 10: No, last digit 5 is not 0 (1).	1+1+1
16	Digit sum of the other digits $= 16$ (1); need a total that is a multiple of 3 (1); digits: 2, 5, 8 (1).	1+1+1
17	16 and 40: common factors give HCF $= 8$ (1.5); LCM $= 80$ (1.5).	1½+1½
	OR: By 2: Yes, last digit 2 is even (1); by 5: No, last digit 2 is not 0 or 5 (1); by 10: No, last digit 2 is not 0 (1).	1+1+1
18	Digit sum of the other digits $= 17$ (1); need a total that is a multiple of 3 (1); digits: 1, 4, 7 (1).	1+1+1
	OR: 15 and 45: common factors give HCF $= 15$ (1.5); LCM $= 45$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	By 2: No, last digit 7 is not even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: No, last digit 7 is not 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: No, last digit 7 is not 0 (1);	1+1+1+1+1
	OR: Remainder when 5603 is divided by 10 is 3 (2); number to add $= 10 - 3 = 7$ (2); new number 5610 (1).	2+2+1
20	By 2: No, last digit 5 is not even (1); By 3: Yes, digit sum 18 is a multiple of 3 (1); By 5: Yes, last digit 5 is 0 or 5 (1); By 9: Yes, digit sum 18 is a multiple of 9 (1); By 10: No, last digit 5 is not 0 (1);	1+1+1+1+1
	OR: Remainder when 3874 is divided by 9 is 4 (2); number to add $= 9 - 4 = 5$ (2); new number 3879 (1).	2+2+1

Mistakes to avoid:

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